

Chapter 21

AI in Social Work Research



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1 Introduction

Artificial intelligence (AI) has evolved from a peripheral innovation into a central methodological component across the social sciences, including social work research. In recent years, scholars have adopted data mining and natural language processing (NLP) to identify social needs in case notes and community narratives. Predictive analytics have been used to model risk and support decision-making for service allocation in high-stakes contexts (Garkisch & Goldkind, 2025). Computer vision and socially assistive robots now enable the observation of social environments and the design of technology-mediated interventions (Hwang & Naik, 2023). Large language models (LLMs) further expand these possibilities by enabling the integration of qualitative and quantitative inquiry. They can assist in coding, synthesis, and hypothesis generation, yet they also raise new concerns about accuracy, bias, and data provenance (Hua et al., 2025). Collectively, these developments create opportunities to analyze complex, multimodal evidence at scale and to generate testable theories on how policies and programs affect individuals and communities.

However, the use of AI in social work also brings forward challenges that are particularly relevant to the profession's human rights mandate. The opacity of

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black-box models complicates interpretability and validity (Reamer, 2023). Training data often reproduces structural inequities (Keddell, 2019), while AI-assisted research raises ethical questions about privacy, consent, and governance when sensitive client information is involved. Evidence from applied systems shows that implementation choices, such as model design, threshold selection, and decision support policies, can shape downstream outcomes and equity. Researchers are thus reminded to evaluate not only statistical performance but also institutional context and patterns of human–AI interaction. These methodological, ethical, and epistemological issues determine whether AI strengthens or undermines social work's commitments to dignity, justice, and transparency.

This chapter examines the current state of AI in social work research with three aims. First, it organizes emerging applications through a research design lens: descriptive and exploratory studies that employ data mining and NLP to map needs and narratives; predictive and analytical studies that develop and interrogate risk models across diverse human service systems; experimental and intervention studies that use socially assistive robots (SARs), computer vision, and conversational agents to test technology-mediated practices; and synthesis and conceptual studies that leverage LLMs, retrieval-augmented generation (RAG), and agentic AI to support theory building and knowledge integration. Second, the chapter identifies cross-cutting methodological, ethical, and epistemological challenges, including integrating AI into mixed-methods designs, pursuing reproducibility and interpretability, and protecting privacy, consent, and fairness. Finally, it reflects on how AI reshapes foundational questions of knowledge production through evolving forms of human–AI collaboration. The goal is to offer a critical, practice-informed roadmap for rigorous, transparent, and equity-centered research in social work.

2 Applications of AI in Social Work Research

The applications of AI in social work research are diverse, spanning multiple methodological traditions and analytic goals. Rather than operating as a one-size-fits-all solution, AI serves as a modular methodological partner that can augment different stages of inquiry, from data collection and pattern discovery to prediction, experimentation, and conceptual synthesis. Its usefulness depends on how well specific techniques are aligned with the epistemic aims and design requirements of a given study. The sections that follow examine how AI methods have been integrated into four major research designs in social work: descriptive and exploratory studies that map needs and narratives; predictive and analytical studies that model risks and complex outcomes; experimental and intervention studies that test technology-mediated practices; and synthesis and conceptual studies that use AI to generate, integrate, and refine theoretical frameworks.

2.1 *Descriptive/Exploratory Research*

Descriptive and exploratory studies in social work aim to observe social patterns, identify salient factors, and generate hypotheses without directly testing causal relationships. Traditional methods, such as surveys, administrative reports, and ethnographic fieldwork, have yielded valuable insights but are often constrained by limited sample sizes, subjective coding, and high labor demands. Artificial intelligence offers scalable tools that can process large, unstructured datasets and uncover latent structures. Among these, three techniques have become especially prominent: text mining, natural language processing (NLP), and causal AI.

2.1.1 Text Mining

Text mining refers to a family of computational techniques that transform unstructured text into analyzable data (Zhang & Li, 2023). Before the recent wave of LLMs, most text mining in social work relied on relatively simple but powerful pipelines that combined basic preprocessing, descriptive statistics, and classical machine learning (Hou & Huang, 2025). These methods remain valuable for descriptive and exploratory research because they are transparent, reproducible, and well aligned with qualitative and mixed-methods traditions.

A typical workflow begins with basic text preprocessing and descriptive counts. Researchers tokenize text into words or sentences, convert all characters to lower-case, remove punctuation, and sometimes remove high-frequency stopwords such as “the” or “and.” Stemming or lemmatization is used to collapse word variants so that “helping,” “helped,” and “helps” are treated as instances of the same concept. From there, analysts compute word and phrase frequencies, often examining the most common unigrams, bigrams, and trigrams to gain an initial sense of what respondents discuss in open-ended survey items, case notes, or online discussions. Keyword-in-context (KWIC) or concordance views extend these counts by showing each occurrence of a target term with a small window of surrounding words, which is crucial for qualitative interpretation (Manning & Schütze, 1999). For instance, it is not only how often “violence” or “stigma” appears that matters, but whether these terms are used to describe personal experience, institutional practices, or public discourse.

To move beyond single words, text mining frequently incorporates co-occurrence analysis, collocations, and simple networks. Co-occurrence tables record which words or codes tend to appear within the same sentence, paragraph, or document (Callon et al., 1991). These tables can reveal, for example, that terms related to “stigma” regularly co-occur with “healthcare,” “police,” or “child welfare,” suggesting relationships among concepts that merit deeper qualitative exploration. Collocation and n-gram analysis (Smadja, 1991) help identify multiword expressions such as “food insecurity,” “case manager,” or “risk assessment” that function

as meaningful units in practice settings. Researchers often visualize these relationships as networks, where nodes represent words or codes and edges reflect the strength of co-occurrence. Clusters of tightly connected terms can surface emergent topics, such as intersecting discussions of housing, mental health, and criminal justice, which are then interpreted using domain knowledge and qualitative reading.

Topic modeling and related unsupervised methods offer another set of tools that became popular in mixed-methods work before generative AI. Latent Dirichlet Allocation (LDA) represents documents as mixtures of a small number of topics, each defined by a probabilistic distribution over words (Kaur & Wallace, 2024). In social work research, LDA has been used to identify candidate themes across large corpora of narrative records, such as thousands of case summaries or social media posts, and to compare topic prevalence across time periods, client groups, or intervention conditions (Benites-Lazaro et al., 2018; Sun et al., 2024; Park, 2024).

Structural topic models extend this approach by allowing document-level covariates, such as gender or service setting, to influence topic prevalence and, in some cases, topic content. Earlier work also employed latent semantic analysis and other matrix factorization techniques to uncover latent dimensions in term-document matrices, which can support clustering, visualization, or exploratory scaling of concepts (Qi et al., 2024b; Yeasin et al., 2009). Dictionary-based sentiment analysis tools, which score texts using predefined lexicons of positive and negative words, have been used to approximate attitudes or emotional tone in narratives about policy changes, service experiences, or public campaigns (Hardeniya & Borikar, 2016).

Once some human coding exists, supervised machine learning methods can be trained to predict codes or categories at scale. These approaches typically represent text segments using bag-of-words or TF-IDF vectors, then fit models such as logistic regression, support vector machines, naive Bayes, or tree-based classifiers (Dawar et al., 2023). In social work contexts, such models have been used to predict whether a sentence or note mentions particular constructs, for example, structural racism (Mohebbi et al., 2023), housing instability (Wilson et al., 2020), or intimate partner violence (Ha et al., 2021), and to prioritize ambiguous or high-risk segments for human review through active learning (Hughes et al., 2021). In systematic reviews (Neubauer et al., 2023; Novitzky et al., 2023) of computational text analysis on domestic and intimate partner violence, these supervised models appear alongside unsupervised methods and rule-based systems, illustrating how relatively simple text mining pipelines can scale qualitative coding to tens or hundreds of thousands of documents while still relying on human judgment for interpretation.

Together, these pre-generative AI text mining techniques provide a bridge between traditional qualitative analysis and more advanced NLP methods. They enable social work researchers to summarize large bodies of narrative data, surface candidate themes and relationships, and generate hypotheses for subsequent in-depth qualitative or quantitative investigation, while maintaining a level of

methodological transparency that supports critical reflection and community engagement.

2.1.2 Natural Language Processing

NLP is a branch of AI that enables computers to understand, generate, and analyze human language. It integrates techniques from linguistics, computer science, and machine learning to process unstructured text at scale. Modern NLP systems combine rule-based methods, statistical modeling, and deep learning architectures such as transformers to capture syntax, semantics, and context. The recent surge in NLP—accelerated by advances in neural networks and the availability of large textual corpora—has enabled sophisticated linguistic analysis with unprecedented accuracy. These capabilities are particularly valuable in social work research, where much of the field's evidence is embedded in qualitative narratives, clinical notes, and open-ended case reports. By automating the extraction and organization of these narratives, NLP enables researchers to uncover patterns in language use, identify emerging social issues, and transform qualitative insights into analyzable data, thereby offering a promising paradigm to bridge the gap between qualitative inquiry and large-scale quantitative analysis.

In applied research, NLP has been used to convert unstructured text into structured datasets suitable for empirical analysis. Bako and colleagues, for example, developed an NLP and machine learning pipeline to classify social work interventions recorded in electronic health record (EHR) notes (Bako et al., 2021). Their algorithm successfully categorized thousands of intervention descriptions, demonstrating the feasibility of automated extraction. The authors argued that such tools could improve workforce planning and facilitate research on the effectiveness of interventions. Another notable application involves the detection of intimate partner violence (IPV) within health records. Tabaie and colleagues compared structured coding with NLP-based approaches. They found that a rule-based NLP algorithm identified IPV-related emergency-department encounters with 99.5% precision by scanning clinical notes for situational cues (Tabaie et al., 2022). They suggested that near-real-time detection could enhance surveillance and enable faster social work intervention.

Beyond clinical data, NLP has also been applied to analyze social media and police reports (An et al., 2023; Yang et al., 2023). Neubauer and colleagues conducted a systematic review that identified 22 studies using computational text analysis to explore domestic and intimate partner violence (Neubauer et al., 2023). These studies employed a range of techniques under the broader NLP umbrella, combining rule-based classification with machine learning and deep learning, as well as topic modeling for pattern discovery. Such approaches integrate language understanding with statistical and neural representations, enabling researchers to detect patterns that traditional manual coding cannot capture. Most datasets were drawn from social media platforms (15 studies), while others originated from police reports, health records, and social service agencies. Typical research tasks included

analyzing the #WhyIStayed campaign to understand reasons for remaining in or leaving abusive relationships and detecting crisis-related posts on Facebook. Together, these studies illustrate how NLP and related AI methods extend text mining capabilities, enabling the systematic exploration of sensitive social issues while underscoring the need for ethical safeguards regarding privacy, consent, and data representativeness.

2.1.3 Causal AI

Causal AI refers to a family of methods that use explicit causal models to move beyond correlation and explore how changes in one part of a system can affect another (Mannava, 2024). Central tools include directed acyclic graphs (DAGs), structural causal models, and potential outcomes frameworks, which allow researchers to encode assumptions about confounders, mediators, and selection mechanisms (Chernozhukov et al., 2024). For descriptive and exploratory social work research, these models are especially valuable because they provide a transparent way to articulate hypothesized causal structures, identify which variables must be measured and adjusted for, and generate testable hypotheses about how policies and programs shape client outcomes.

In practice, DAGs and related causal diagrams can be used to map complex service systems, such as child welfare, homelessness prevention, or integrated behavioral health care. By visually representing relationships among family characteristics, neighborhood conditions, prior system involvement, and service decisions, social work researchers can distinguish between confounders that should be controlled, colliders that should not be conditioned on, and mediators that may carry the effect of an intervention (Austin et al., 2019; Poppe et al., 2025). These diagrams help clarify whether an observed association, such as between screening scores and placement decisions, is plausibly causal or likely to be confounded by unmeasured structural factors. They also inform data collection and model specification by highlighting which variables are essential to include before conducting predictive or evaluative analyses.

Causal AI also encompasses algorithmic approaches that combine machine learning with causal inference to estimate treatment effects and identify heterogeneous impacts across subgroups. Methods such as causal forests, targeted maximum likelihood estimation, and double machine learning can use high-dimensional administrative and narrative data to estimate how specific interventions, such as in-home services versus foster care placement, supportive housing versus usual shelter use, or chatbot-based psychoeducation versus standard information, differentially affect outcomes for clients with distinct profiles (Lechner & Mareckova, 2024; McConnell & Lindner, 2019). These models remain exploratory when used to generate hypotheses about which families or communities are most likely to benefit from a given program. Still, they can guide subsequent confirmatory studies and more rigorous impact evaluations.

Finally, causal discovery techniques use statistical constraints to search for plausible DAG structures directly from data under explicit assumptions. Although still

emerging in social work, these methods may help researchers uncover candidate causal pathways in complex systems where theory is underdeveloped, such as the interplay among online misinformation, help-seeking behavior, and service engagement. For example, Tian and Rizoïu (2025) used a structural causal modeling and counterfactual framework to estimate how public attention signals, such as Google search trends, causally shape engagement of misinformation. Their findings show that temporal shifts in information-seeking behavior produce measurable causal effects on online participation, illustrating how causal AI can reveal influence pathways relevant to help-seeking and service engagement. Because causal AI makes assumptions explicit and invites critical scrutiny, it fits well with participatory and community-engaged research traditions. When combined with qualitative inquiry and practitioner knowledge, causal AI can support more transparent theorizing about mechanisms of change and more thoughtful design of interventions that target leverage points within social systems.

2.2 Predictive/Analytical Research

Predictive and analytical research in social work uses AI to anticipate future outcomes, characterize risk and resilience profiles, and identify leverage points for intervention. Early work focused on relatively transparent models such as regularized regression and tree-based ensembles built from structured administrative data (Shortreed et al., 2023; Khan et al., 2024b). Over time, however, the methodological repertoire has expanded to include unsupervised learning, dimensionality reduction, multimodal prediction, and forecasting models, as well as privacy-preserving training schemes such as federated learning (Che et al., 2023; Nikolaidis et al., 2023). These approaches allow researchers to discover natural groupings of clients, uncover latent dimensions of need, combine text, images, and sensor streams, and generate short- and long-term forecasts of service demand or risk trajectories. Together, they broaden the ways predictive analytics can inform social work practice, moving beyond single-outcome risk scores toward richer descriptions of patterns, pathways, and potential intervention touchpoints.

2.2.1 Regularized Regression Models and Structured Predictors

Regularized regression approaches, such as LASSO, ridge regression, and elastic net, remain widely used in social service analytics due to their interpretability and ability to handle large predictor sets. One influential example is the Allegheny Family Screening Tool (AFST), which employs regularized logistic regression to estimate the likelihood of adverse child welfare outcomes using hundreds of administrative and demographic variables (Allegheny County Department of Human Services, Office of Analytics, Technology and Planning, 2024). Trained on historical referral data and periodically recalibrated, AFST has been shown to reduce unnecessary investigations for low-risk referrals, improve decision consistency at

higher risk levels, and be associated with reduced racial disparities at some decision points. Similarly, Rosholm et al. (2024) analyzed nearly 200,000 child maltreatment referrals in Denmark using logistic regression and LASSO, reporting AUROC values above 0.87 and demonstrating strong correlations between predicted risk and later adverse outcomes. These structured models illustrate how regularized regression can support prioritization, triage, and early-warning analytics while maintaining statistical transparency.

2.2.2 Tree-Based Ensemble Methods

Tree-based ensemble models, such as random forests and gradient boosting, offer a more flexible alternative for capturing nonlinear relationships and complex interactions among risk factors. They have been particularly impactful in homelessness and housing instability research. Byrne et al. (2019) applied random forests to electronic medical records from more than 5.8 million U.S. veterans, achieving substantially higher sensitivity than logistic regression in predicting positive homelessness screens, at the cost of some reduction in specificity. Building on this, Zamora-Resendiz et al. (2023) integrated document features and structured EMR data to predict homelessness risk months before formal screening, with the top 1% of model-identified individuals being 34 times more likely to screen positive than a random selection. Such results demonstrate that ensemble methods offer superior early-detection capability and can uncover latent high-risk groups missed by traditional screening.

2.2.3 Multi-model Decision Support Systems and Placement Optimization

More recent work has explored multi-model systems that combine machine learning predictions with placement decision algorithms. Trudeau et al. (2023) developed 50 random-forest models to predict successful placement for youths entering psychiatric residential treatment or community care, achieving AUROCs of 0.66–0.79. In validation analyses, 80% of youths matched to model-recommended placements exceeded the predicted outcome, suggesting strong alignment between algorithmic recommendations and real-world trajectories. Beyond youth treatment settings, multi-model placement systems incorporate diverse predictors—clinical histories, behavioral indicators, prior service use, and contextual variables—to generate individualized risk-benefit profiles that augment professional judgment. These systems help reduce reliance on subjective heuristics, promote equity in resource allocation, and support multidisciplinary teams in allocating scarce high-intensity services.

2.2.4 Unsupervised, Multimodal, and Privacy-Preserving Predictive Models

In addition to supervised prediction of specific outcomes, social work researchers increasingly use unsupervised learning to explore heterogeneity in client populations and service trajectories (Annaki et al., 2022). Cluster analysis, for example, Fink (2025) applied k-means, Gaussian mixture models, and hierarchical clustering to identify subgroups of individuals who shared similar behavioral and demographic patterns, demonstrating how unsupervised methods can reveal latent structures across many variables within complex social data. Studies have applied these methods to identify distinct profiles of child welfare involvement, combinations of housing instability, mental health symptoms, and justice contact, or trajectories of substance use and treatment engagement (Putnam-Hornstein et al., 2013). Rather than predicting a single binary outcome, clustering reveals naturally occurring subgroups (Chin & Collins, 2024), such as families with high material hardship but low formal system contact, that may require different outreach strategies or policy responses. Mixture models and trajectory clustering extend this logic to longitudinal data, grouping individuals by their patterns of change over time rather than only by their status at a single point in time.

Dimensionality reduction methods such as principal component analysis (PCA) and related factor models provide another set of tools for analytical research. When administrative datasets contain dozens or hundreds of correlated indicators, PCA can summarize them into a smaller number of composite indices that capture underlying constructs such as neighborhood disadvantage, service complexity, or cumulative adversity (Friesen et al., 2016). These latent dimensions can then be used as predictors in outcome models or as grouping variables to examine how intervention effects vary across contexts. In recent work, autoencoder-based representation learning plays a similar role, compressing high-dimensional records or free-text embeddings into dense feature vectors that preserve substantial variation while reducing noise (Kane et al., 2023).

Multimodal predictive models integrate heterogeneous data sources that increasingly characterize social work practice, including structured records, narrative notes, images, audio, and sensor data. For instance, models may combine case characteristics, NLP-processed worker notes, and neighborhood indicators to forecast referrals to child protective services, or they may integrate wearable sensor streams with self-reported mood to predict crisis episodes in community mental health (Kline et al., 2022; Saha et al., 2021). Sequence models and other time-aware architectures can learn temporal patterns in these multimodal streams, improving forecasts of short-term risk or service utilization. For example, Othman et al. (2024) used a multimodal time-series framework combining social media emotion signals and hospital emergency data to forecast public mental health needs during the COVID-19 pandemic in Singapore, showing how external text-derived indicators can anticipate surges in crisis visits. At the system level, Megouo and Pierre (2024) developed an ensemble forecasting model that accurately predicted emergency medical call volumes by integrating temporal, spatial, and environmental features,

demonstrating how forecasting methods can support proactive planning of staff and resources across human service agencies.

Finally, federated learning and related privacy-preserving techniques address the reality that much social work data is fragmented across agencies that cannot easily pool records due to legal, ethical, or logistical constraints. In a federated learning setup, models are trained across multiple organizations or jurisdictions without sharing raw data. Each site computes model updates locally, and only encrypted summaries are aggregated, enabling the development of more generalizable risk models or resource allocation tools while limiting data transfer (Loftus et al., 2022). Although still emerging in social work, these approaches are being explored in allied fields such as healthcare and education, and they hold promise for cross-agency predictive analytics in areas such as homelessness prevention, child welfare, and intimate partner violence services (Abbas et al., 2024). At the same time, they raise new questions about governance, equity, and accountability when decision support tools are built from data that no single stakeholder can easily inspect (Eden et al., 2025).

2.3 Experimental/Intervention Research

Experimental and intervention research leverages AI-driven technologies to test innovative methods to enhance client well-being, strengthen practitioner competencies, and improve the delivery of social services, unlike predictive analytics, which primarily model risk and outcomes. Experimental approaches use AI as an active component of the intervention itself. A growing set of technologies, such as socially assistive robots, conversational agents, and computer-vision-enabled tools, is increasingly being deployed to deliver therapeutic interactions, provide skills training, monitor behavioral change, and augment frontline practice. These tools are particularly well-suited to social work because they support consistent, scalable, and context-sensitive engagement with clients while addressing longstanding challenges in the field, including workforce shortages, variability in service quality, and the need for culturally responsive and accessible interventions.

2.3.1 Socially Assistive Robots

Socially assistive robots (SARs) integrate AI-based perception, language processing, and adaptive decision-making to provide support through social interaction rather than physical labor (Yang et al., 2025). Contemporary SARs rely on natural language understanding, speech generation, emotion recognition algorithms, and behavior planning models to sustain conversation, respond to user affect, prompt activities, and deliver tailored feedback. These technical affordances make SARs well-suited for experimental research in social work, where investigators can systematically vary the robot's dialogue style, emotional responsiveness, or

prompting strategy to test their causal effects on client engagement, psychological well-being, or skill acquisition. A scoping review of SARs in mental health care identifies two primary forms used in such studies: Companion or animal-like robots, which emulate features of animal-assisted therapy without the risks of allergens or care burden; and humanoid or service-oriented robots, which support more natural verbal and nonverbal interaction and can autonomously deliver structured activities (Guemghar et al., 2022). Early experiments show that SARs can reduce distress among hospitalized children (Trost et al., 2019), model social behaviors for youth with autism (Cabibihan et al., 2013), and support exercise adherence and health monitoring among older adults (Gadde et al., 2011). However, empirical evidence remains limited by small samples and narrow diagnostic coverage.

A recent randomized controlled trial by Murayama and Takase (2025) demonstrates how SARs function as an experimental intervention tool. Community-dwelling older adults in Japan were assigned either to a humanoid robot intervention—featuring AI-driven conversation, reminders, and cognitive games—or to care-as-usual. Relative to controls, the robot group reported significant reductions in loneliness and improvements in psychological well-being. Qualitative responses highlighted the robot’s perceived emotional support, companionship, and cognitive stimulation. Notably, the study illustrates how SARs enable researchers to isolate the effects of specific communicative or behavioral modules (e.g., empathic responses, motivational prompts) delivered consistently across participants—capabilities that are difficult to achieve with human facilitators. Thus, SARs represent a promising platform for controlled experimentation in social work research, allowing investigators to test theoretically informed mechanisms of social support, engagement, and behavior change within highly standardized yet socially meaningful interactions.

2.3.2 Conversational Agents

Modern conversational agents (CAs) increasingly rely on advanced AI capabilities—including LLMs, context-aware dialogue management, reinforcement learning-based adaptation, and agentic planning systems—to deliver interactive, human-like communication. Unlike earlier rule-based or decision-tree systems, LLM-enabled CAs can interpret open-ended user input, generate personalized responses, and sustain emotionally attuned dialogue. These technologies are highly valuable in experimental social work research because they enable investigators to manipulate variables such as empathic style, motivational strategies, cultural tailoring, or fidelity to therapeutic protocols in a controlled yet scalable manner. In this sense, CAs function not simply as service tools but as experimental instruments: they can standardize intervention delivery, model specific communication mechanisms, and simulate practitioner–client interactions for training, mechanism testing, and causal inference.

A growing body of evidence demonstrates the potential and limitations of these systems. A 2023 meta-analysis of 32 randomized controlled trials found that CA interventions produced modest but significant short-term improvements in depressive symptoms, generalized and specific anxiety, distress, stress, and quality of life, with personalization and empathic responsiveness identified as key active ingredients (He et al., 2023). Related scoping reviews show that AI chatbots used for professional training, often combining elements of cognitive-behavioral therapy and scenario-based simulation, can reduce anxiety and burnout among health and social care workers. However, the evidence remains sparse (Baek et al., 2025). Although these studies primarily examine CAs as therapeutic or training interventions, they also demonstrate that CAs can reliably deliver communicative components, such as empathic responding, motivational prompts, and structured feedback, consistently across participants. This reliability is a necessary methodological condition for using CAs as experimental instruments. For social work research, these findings are significant: CAs allow researchers to experimentally test the effects of communication style, feedback mechanisms, and relational behaviors on client engagement and practitioner skill development; evaluate the scalability and consistency of intervention protocols; and investigate mechanisms of change within digitally mediated helping relationships. As such, AI-powered conversational systems serve as versatile research platforms for probing, refining, and experimentally validating interventions in social work.

2.3.3 Computer Vision-Enabled Interventions

Computer vision refers to methods that allow computers to interpret information from images and video, such as detecting objects, estimating body pose, or recognizing patterns in the built environment (Mendes-Neves et al., 2023). In social work research, computer vision-enabled interventions typically use cameras or other imaging devices to monitor risk, support safety, and augment practitioner judgment in real time (Bernasco et al., 2021; Prakash et al., 2023). These systems do not replace social workers or caregivers. Instead, they provide additional streams of information that can prompt timely follow-up, inform assessment, and guide resource allocation.

A critical application area involves safety monitoring and risk detection in settings such as long-term care facilities, domestic violence shelters, and supported housing (Mantler et al., 2021; Ullah et al., 2025). For example, vision models can be trained to detect falls, wandering, or prolonged inactivity among older adults who live alone or in assisted living. Chen and Fisher (2023) developed the MISO system, which uses RGB-D cameras to monitor inactivity and automatically alert caregivers when it detects prolonged stillness or a possible fall, allowing staff to check in immediately rather than relying on periodic in-person rounds or self-reports. Similar approaches have been explored for detecting aggressive incidents or escalating conflict in residential group homes, where automatic alerts can trigger de-escalation protocols or call additional staff to the scene (Thomas & Balamurugan,

2024). In these applications, computer vision functions as an always-on observer that can recognize predefined risk signatures and route cases to human decision-makers.

A second line of work uses computer vision to assess physical and neighborhood environments that shape social work concerns. Researchers have analyzed street view images, satellite imagery, and building photographs to quantify features such as housing conditions, sidewalk quality, the presence of green space, and signs of physical disorder (Larkin et al., 2025; Suel et al., 2021). These image-based indicators can be linked to administrative data to examine how environmental characteristics relate to crime, child maltreatment reports, chronic disease, and service utilization (Wu & Wang, 2024; Yue et al., 2022). For instance, image-based measures of housing disrepair or neighborhood blight can help identify blocks where housing inspection, lead abatement, or community outreach programs may be most needed. Ercan and Mathyalakan (2024) developed a computer vision-based system that automatically inspects public housing images to detect safety risks such as clutter, obstruction, or structural damage, enabling authorities to prioritize inspections and respond more quickly to potential hazards. In this way, computer vision supports place-based interventions by turning visual cues into structured data that can be mapped and tracked over time.

Computer vision has also been used to support practice and training by analyzing video-recorded interactions. In child welfare, researchers have explored tools that estimate synchrony between caregivers and children during play sessions by tracking gaze, proximity, and body posture (Isaev et al., 2023; Weng et al., 2025). In mental health and school-based counseling, computer vision models have been piloted to detect basic nonverbal cues, such as eye contact, head nods, and extended silences, during simulated sessions (Inoue et al., 2021; Vail et al., 2021). For example, Choi et al. (2024) developed a virtual counseling system that used computer vision to model nonverbal compassion behaviors—such as gaze alignment, facial mimicry, and head-nodding—and found that these cues improved perceived empathy and communication effectiveness in counseling simulations. These signals can be summarized for supervision, providing students and practitioners with structured feedback on engagement patterns that would be difficult to track manually across many sessions. Although such systems are not used to evaluate performance in a high-stakes way, they illustrate how vision-enabled analytics can serve as mirrors that help practitioners reflect on their own practice.

Finally, several computer vision applications aim to promote accessibility and inclusion for people with disabilities, which is a central concern in many social work settings. Mobile applications that recognize printed text, signage, or everyday objects can help people with visual impairments navigate public spaces and service environments (Chimwaza & Jimu, 2024). Sign language recognition and gesture-based interfaces can facilitate communication between deaf or hard-of-hearing clients and providers who do not share a common signed language (Alsharif et al., 2025). In child and adolescent services, vision-based tools have been used experimentally to support social skills training. For example, in the ALTRIRAS system developed by Almeida et al. (2019), which uses real-time facial expression

recognition to give children immediate visual feedback during structured emotion recognition games, helping them practice identifying and responding to basic emotions. These interventions blend human support with automated cues to lower barriers to participation and communication.

Together, these examples show that computer vision-enabled interventions can extend social work practice beyond traditional text and tabular data, bringing visual information about bodies, spaces, and interactions into analytic and decision-making systems. They also set the stage for critical ethical questions about surveillance, consent, and potential misuse, including applications such as targeted policing and facial recognition, which are taken up in more detail in later sections on equity and governance.

2.4 Synthesis/Conceptual Research

Synthesis and conceptual research in social work examines how AI systems can generate new knowledge, integrate complex evidence, and contribute to theory building. Unlike descriptive or exploratory analyses that characterize patterns in existing data, this strand of research uses AI as an intellectual instrument: a tool for organizing large bodies of information, surfacing latent patterns, articulating conceptual frameworks, and interrogating underlying assumptions about human behavior and social systems (Alam, 2025; Mökander & Schroeder, 2021). Recent scholarship argues that LLMs can function as synthetic theorists, capable of scanning disciplinary corpora, mapping conceptual spaces, and proposing candidate mechanisms for subsequent evaluation (Zhang et al., 2025). Experimental work in scientific discovery platforms further shows that LLMs can integrate heterogeneous evidence sources to generate plausible hypotheses and identify unrecognized linkages across literatures, suggesting that such systems may assist researchers in developing new theoretical propositions (Jiang et al., 2025; Qi et al., 2024a).

The advantages of AI-driven synthesis are especially salient in a field like social work, where knowledge is fragmented across disciplines and where theories often integrate psychological, relational, structural, and policy-level determinants. Studies evaluating LLM-supported evidence synthesis have shown that models can reconcile conflicting findings, trace multidimensional mechanisms, and generate structured summaries that approximate expert reasoning, thereby supporting conceptual integration across domains (Dennstädt et al., 2024). In related work, multi-agent LLM systems have been used to orchestrate multi-step conceptual tasks such as generating alternative framings, evaluating competing logic models, and constructing causal diagrams from complex policy or program documents (Alba et al., 2025). Through these capabilities—rapid evidence aggregation, counterfactual reasoning, and iterative conceptual refinement—AI serves not merely as a summarization tool but as a collaborator for conceptual innovation, complementing human judgment and extending the methodological repertoire available for theory-driven social work scholarship.

2.4.1 Large Language Model-Based Conceptual Synthesis and Theory Development

LLMs offer powerful capabilities for synthesis and conceptual development because they can process large, interdisciplinary evidence bases and generate structured insights that traditionally required extensive expert labor. In conceptual and theory-building research, particularly in fields such as social work, where explanations span multiple levels of analysis, LLMs enable researchers to rapidly map conceptual linkages, compare competing frameworks, surface latent constructs, and propose theoretical extensions that can later be empirically tested. Compared to manual qualitative synthesis or expert-driven literature reviews, LLMs provide scale, speed, and the ability to integrate heterogeneous sources, reducing human bias while enabling iterative refinement of conceptual models.

For example, a recent study on literature screening workflows demonstrated that LLMs can be used to evaluate the relevance of scientific publications based on title and abstract, achieving high sensitivity in identifying eligible studies for systematic reviews while substantially reducing the number of records that require manual screening (Dennstädt et al., 2024). In that work, the model was prompted to apply prespecified inclusion criteria, assign a relevance score, and generate short justifications, allowing human reviewers to focus on borderline cases and audit the reasoning behind inclusion or exclusion decisions. This type of application illustrates how LLMs can participate in conceptual synthesis not only by summarizing existing studies, but also by shaping the evidence base from which theories and practice guidelines are derived. Another perspective article argues that LLMs are transforming the scientific method, from hypothesis generation through discovery, by enabling automated reasoning across large disciplinary corpora (Zhang et al., 2025). The authors describe pipelines in which LLMs map existing constructs, identify gaps or contradictions in current theories, suggest novel hypotheses, and even propose candidate study designs or measurement strategies that can then be refined and evaluated by human researchers. Finally, in computational social science, a review of LLM applications shows that these models are already being used to generate structured insights from heterogeneous text datasets, supporting tasks such as stance detection, social network inference, and topic modeling in social system research (Thapa et al., 2025). In that literature, LLMs have been used to code frames in protest speech, infer relational ties from co-mentioned actors in news coverage, and derive high-level themes from extensive collections of policy documents, thereby providing building blocks for new theories of polarization, mobilization, and institutional change.

2.4.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) frameworks (Gao et al., 2023) are well-suited to social work synthesis and conceptual research because they combine the generative reasoning of LLMs with the evidentiary grounding of retrieval systems.

This allows researchers to produce concept maps, cross-study comparisons, policy analyses, and theoretical arguments that are explicitly tied to cited empirical sources. Whereas traditional LLM-based synthesis may rely on latent statistical patterns learned during pretraining, RAG can systematically retrieve, summarize, and contrast large volumes of research, reducing hallucination and increasing transparency by grounding outputs in verifiable documents.

In healthcare, RAG has been used to reduce factual errors and produce trustworthy, evidence-linked responses. For example, a recent COVID-19 fact-checking study integrating GPT-4 with multiple RAG variants demonstrated substantial gains in accuracy and significant reductions in hallucinations when verifying claims against a corpus of 130,000 peer-reviewed articles, highlighting the value of retrieval grounding for high-stakes scientific synthesis (Li et al., 2025). Building on this logic, a 2024 ACL paper showed that RAG-generated explanations grounded in local news articles about socioeconomic problems were rated higher than standard LLM outputs on factual accuracy, completeness, and contextual relevance. In that work, the system retrieved local reporting about housing, employment, and social programs, then used those documents to generate explanations of neighborhood-level challenges and policy options, illustrating how retrieval grounding can anchor abstract reasoning in place-based evidence (Dumitru et al., 2024).

Emerging applications in social and human services further illustrate the breadth of issues RAG can address. Pilot systems for homelessness services have used RAG to answer questions about shelter availability, eligibility criteria, and local outreach programs by retrieving information from municipal service directories, agency websites, and community resource guides, while linking users back to the underlying sources (Khan et al., 2024a; Singh et al., 2025). In child welfare and family services, prototype tools retrieve policy manuals, case law summaries, and prior assessment reports when workers ask questions about placement options, visitation rules, or mandated timelines, helping align decisions with current regulations and evidence without requiring staff to search multiple databases manually (Wu et al., 2025). In public health and social policy communication, experimental RAG assistants have been designed to respond to questions about social benefits, vaccine safety, and stigma-sensitive topics by retrieving passages from guidelines, systematic reviews, and advocacy reports, then generating tailored explanations that preserve citations and highlight conflicting or conditional findings (Amugongo et al., 2025). Together, these examples suggest that RAG could support social work researchers and practitioners in creating comprehensive evidence syntheses, exploring contradictory findings across disciplines, and generating contextually grounded policy and practice narratives explicitly tied to verifiable documents.

2.4.3 Agentic AI

Agentic AI extends beyond static text generation by enabling multi-step reasoning, planning, tool use, and autonomous execution of research tasks (Ali & Dornaika, 2025). For synthesis and conceptual research in social work, agentic systems offer unique advantages: they can orchestrate literature searches, evaluate the quality of evidence, compare theoretical models, generate alternative conceptual framings, and iteratively refine hypotheses in response to researcher feedback. This workflow-driven autonomy distinguishes agentic AI not only from traditional qualitative synthesis but also from standard LLM or RAG outputs, enabling more systematic exploration of complex social phenomena and multi-level theories.

Emerging evidence shows that agentic AI can support resource-intensive qualitative synthesis tasks (Lee et al., 2025; Xu et al., 2025a). One study evaluating GPT-based agents for thematic analysis demonstrated that carefully designed, stepwise prompting workflows could autonomously extract, synthesize, and justify study-level themes from qualitative interview transcripts. Using a 5-point evaluation scale across 11 published studies, GPT-generated themes achieved an average score of 3.05. The outputs showed strong alignment with researcher-identified themes, high succinctness with minimal redundancy, substantial explanatory depth, and relevant supporting quotations (Yang et al., 2024a). These findings suggest that agentic AI can substantially reduce the manual labor required for qualitative analysis while still producing evaluable, auditable outputs.

Complementary evidence comes from recent work using multi-agent LLM systems to synthesize institutional-level documents. For example, one study analyzing AI policies across the top 50 U.S. universities employed an iterative prompt chaining and self-refinement framework to identify core themes across dozens of policy documents (Alba et al., 2025). The agentic workflow consisted of multiple LLM agents assigned to document summarization, theme proposal, redundancy reduction, and final theme consolidation. This process uncovered seven cross-cutting themes spanning academic integrity, responsible AI use, data privacy, ethical safeguards, and institutional adaptation. The study demonstrates that agentic AI can coordinate multi-step reasoning across large, heterogeneous corpora, producing policy-relevant insights in contexts where manual synthesis would be prohibitively time-consuming.

Although applications directly targeting social work issues remain nascent, several adjacent studies illustrate how agentic AI can be adapted to social service contexts. Multi-agent LLM pipelines have been used to generate and refine causal diagrams, logic models, and program theories for public health and social policy interventions by assigning different agents the roles of theorist, critic, and synthesizer (Dawid et al., 2025; Le et al., 2024). Other human services pilot projects have leveraged agent workflows to map service eligibility rules across fragmented program documents, reconcile conflicting guidelines, and generate client-specific navigation summaries for housing, nutrition, or benefits programs (Barron et al., 2025). In community health research, agentic systems have been used to iteratively cluster, re-cluster, and validate themes across thousands of open-ended responses concern-

ing stigma, barriers to care, and vaccine misinformation, reducing the cognitive burden on researchers while allowing rapid sensitivity analyses across alternative theoretical framings (Gullahorn et al., 2024; Meng et al., 2025). Together, these emerging directions suggest that agentic AI can support social work scholarship by automating early-stage synthesis tasks, generating competing conceptual explanations, and enabling researchers to focus on higher-order interpretation, critique, and integrative theory building.

3 Methodological, Ethical, and Epistemological Challenges

As AI systems become increasingly integrated into social work research, they introduce not only new analytical and conceptual possibilities but also significant methodological, ethical, and epistemological challenges. Unlike conventional tools, AI models, particularly machine learning, LLMs, and agentic systems, shape the research process itself: they influence what can be observed, how constructs are defined, which forms of evidence are prioritized, and how knowledge claims are generated (Barelli et al., 2024; Keyes & Creel, 2022; Lewis et al., 2024). These technologies bring issues such as data bias, opacity, reproducibility, and validity to the forefront of social work scholarship, raising questions about whether AI-enabled findings can be trusted, generalized, or interpreted in ways that remain aligned with social work's values and commitments to equity. At the same time, the scale and automation offered by AI risk amplify existing structural inequalities or embed unexamined assumptions into research designs and theoretical frameworks.

3.1 Methodological Challenges

3.1.1 Integration into Social Research Design

Social work research often relies on small-scale qualitative inquiry, community-based participatory methods, and mixed-methods designs. While AI can extend these approaches by processing large administrative datasets or synthesizing extensive narrative records, effective integration requires attention to alignment with established social research principles at the design stage. Robust frameworks emphasize that AI tools must be embedded into research workflows through transparent protocols for data governance, de-identification, controlled access, baseline model testing, and expert validation. Simply inserting black-box models into qualitative or community-based studies risks privacy breaches, loss of contextual nuance, and unreliable conclusions. Implementation studies in social services further highlight the importance of documenting who uses the AI system, where it is deployed, and for what purpose, as well as developing detailed operational manuals and iterative trial cycles (Nair et al., 2025). These experiences underscore that integration is

not merely a technical task: it requires collaborative planning with practitioners and communities, clear articulation of human-AI roles, and research designs that preserve the epistemic values of social work.

3.1.2 Reproducibility, Interpretability, and Validity

Reproducibility and interpretability remain central concerns in AI-supported studies. Many high-performing machine learning and deep learning models operate as opaque systems, making it difficult for researchers to replicate findings or understand how predictions are generated (Takaoka, 2022). To address these limitations, methodological guidance increasingly recommends incorporating interpretable or explainable techniques, such as linear models, decision trees, Shapley additive explanations, and surrogate models, into the analytic pipeline (Basagaoglu et al., 2022; Nambiar et al., 2023). These approaches help clarify the relationship between inputs and outputs (interpretability) and the reasoning behind specific predictions (explainability). Broader research-integrity discussions reinforce that AI-assisted studies must include transparent methodological descriptions (Chen et al., 2024a), detailed reporting of data preprocessing, and explicit justification of model choices. Taken together, these considerations highlight that validity in AI-supported social work research requires open algorithms where possible, standardized evaluation metrics (e.g., ROC and precision-recall curves), robust documentation, and opportunities for independent replication.

3.1.3 Interpretable and Explainable AI (XAI)

Interpretable and explainable AI, often referred to as XAI, has emerged as a primary subdomain that seeks to address the challenges posed by opaque, black-box systems. Rather than focusing solely on predictive accuracy, XAI methods aim to make model behavior more transparent so that researchers, practitioners, and communities can understand how predictions are generated, identify potential sources of bias, and evaluate whether model reasoning is acceptable for high-stakes decisions (Ortigossa et al., 2024). These goals are especially salient in social work, where AI tools may influence access to services, surveillance, and allocation of scarce resources in ways that directly affect vulnerable populations.

One strand of XAI emphasizes inherently interpretable models whose internal structure can be examined directly. Examples include sparse linear models, decision trees, rule-based learners, and constrained gradient boosting models (Behzadimanesh et al., 2025; Yang et al., 2024b). In child welfare and youth justice research, such models have been used to identify combinations of factors, such as prior reports, caregiver characteristics, school engagement, and neighborhood disadvantage, that precede substantiated maltreatment or system contact (Moon et al., 2024; Moreau

et al., 2024). Because decision paths and feature weights can be inspected and discussed, practitioners can evaluate whether the model relies on meaningful mechanisms or on proxies that may encode structural inequities. Similar approaches have been used in homelessness and housing services to clarify which patterns of shelter use, eviction history, and health conditions are associated with repeated episodes of homelessness, allowing agencies to scrutinize and revise model-based prioritization criteria (Rahmattalabi et al., 2022; Vajiac et al., 2024). For example, Kube et al. (2023) developed a machine learning system that used shelter histories, eviction records, and health information to identify who was most likely to reenter homelessness. By examining which features drove predictions, agencies could adjust their prioritization rules to balance efficiency with equity and ensure that models did not reinforce existing disparities.

A second strand focuses on post hoc explanations for more complex models, including deep neural networks and large ensembles. Techniques such as Shapley additive explanations, Integrated Gradients, Local Interpretable Model Agnostic Explanations, and counterfactual explanations help clarify how changes in specific inputs influence model outputs (Jin et al., 2023; Salih et al., 2023). In predictive analytics for mental health crisis services, for example, Shapley value-based analyses have been used to examine which features, such as recent emergency visits, missed appointments, or neighborhood-level indicators, contribute most strongly to predictions of near-term crisis risk (Kharrat et al., 2024; Mustapha et al., 2022). Counterfactual explanations can show how a small set of modifiable variables would need to change for a client to move from a high-risk to a lower-risk category or from ineligible to eligible for a program, highlighting levers that practitioners or policymakers can realistically influence (Ahmed et al., 2025).

Emerging work also extends XAI to the group level and structural explanations that are critical for equity-oriented social work. Rather than only examining individual predictions, researchers analyze whether feature attributions and error rates differ across demographic groups, jurisdictions, or agencies (Papanikou et al., 2025; Wang & Wu, 2024). In child protection risk assessment, for instance, XAI tools can reveal whether models systematically assign higher risk scores to families from particular neighborhoods or racial groups even when case characteristics are similar (Keddell, 2019; Papanikou et al., 2025). These findings can inform governance processes, community oversight, and model redesign, for example, by adjusting feature sets, changing loss functions, or constraining how certain variables are used. In this way, XAI becomes a mechanism for auditing and correcting AI systems rather than merely describing them.

Integrating XAI into AI-supported social work research strengthens both methodological rigor and ethical accountability. Interpretable and explainable methods make it easier to document model logic, engage practitioners and community stakeholders in understanding and contesting predictions, and align analytic pipelines with social work values. When combined with transparent reporting and independent replication (Gyevnar et al., 2023), XAI offers a practical pathway for ensuring that AI-augmented decisions are not only accurate but also understandable, contestable, and fair for the populations that social work aims to serve.

3.1.4 Data Quality and Representativeness

The performance and trustworthiness of AI techniques depend heavily on the quality of the data used to train them. Social work datasets often contain small sample sizes, inconsistent documentation, and unstructured narrative text—all of which introduce bias and limit model generalizability. Narrative records, for instance, may vary widely in writing style or omit critical contextual information, restricting a model's ability to learn nuanced patterns (Lukmanjaya et al., 2025). Exclusive reliance on administrative data can likewise misrepresent specific populations and reproduce structural inequities. Ensuring representativeness, therefore, requires combining multiple data sources and adopting culturally informed data practices. It also depends on engaging communities, particularly Indigenous populations and other historically marginalized groups, in decisions about data use. Another emerging methodological risk is anchoring bias, in which researchers may over-weight an AI system's initial output because it appears precise or objective (Carter & Liu, 2025) and overlook contradictory evidence. To counter this tendency, AI-generated results must be approached critically, triangulated across sources, and interpreted through human expertise. These concerns collectively highlight the need for rigorous data curation, reflexive engagement with AI outputs, and conservative claims about what AI-derived insights can validly support.

3.2 Ethical Challenges

3.2.1 Privacy, Consent, and Data Governance

AI-assisted social work research often involves highly sensitive information about individuals and communities, raising significant obligations regarding privacy, consent, and data stewardship. Ethical guidance in the field consistently emphasizes that informed consent must be meaningful in technology-mediated settings: participants should understand how their data will be used, what AI systems will access, and what risks, including secondary use and re-identification, may occur. Computer vision applications that rely on continuous video capture, such as fall detection or environmental monitoring, introduce additional challenges because cameras can record bystanders, staff, or residents who have not explicitly agreed to be monitored and whose images can be repurposed in ways they did not anticipate (Moore et al., 2024). These risks are heightened when footage is retained for long periods, combined with other datasets, or shared with external entities, for example, law enforcement agencies or private vendors, potentially enabling forms of targeted surveillance that go well beyond the original care or safety purpose. Robust data governance practices are essential, including strong encryption, secure storage, restricted access protocols, and clear procedures for responding to data breaches. De-identification of narrative records and administrative files is particularly important in domains such as child protection, where rich textual

descriptions may inadvertently reveal identities even when obvious identifiers are removed.

Implementation studies in social services demonstrate how these principles can be operationalized (Trinkley et al., 2024), for example, by using closed networks that prevent external access, allowing workers to enter and review information without exposing it to the public Internet (NASW et al., 2017). These approaches highlight that safeguarding privacy is not only a technical matter but also an ethical responsibility that requires transparent communication with participants, organizational policies for breach notification, and ongoing evaluation of data practices as AI systems evolve. Ethical AI research in social work, therefore, demands a combination of procedural rigor, contextual sensitivity, and continuous human oversight to ensure that data use aligns with social work values and community expectations.

3.2.2 Algorithmic Fairness and Equity

AI systems trained on historical data risk reproducing the very inequities that social work seeks to address. When administrative datasets reflect structural racism, class-based disparities, or biased service patterns, predictive models may encode these patterns and treat them as neutral signals. This can produce disproportionate false positives for marginalized groups, reinforce harmful stereotypes, or create feedback loops in which increased surveillance leads to higher recorded risk (Chapman et al., 2022). Analyses of child welfare risk assessment tools illustrate this concern: algorithms often rely on proxies such as previous system involvement, which correlate with socioeconomic disadvantage and can heighten risk scores for families already subject to disproportionate state attention (Stapleton, 2025). Similar concerns arise with computer vision systems deployed for public safety, benefits administration, or institutional security. Facial recognition technologies have repeatedly been shown to perform less accurately for people of color and other marginalized groups, raising the risk of misidentification, unwarranted investigation, or denial of services when such tools are used around public housing, schools, or welfare offices (Amador-Lankster, 2025; Yucer et al., 2023). Vision systems that purport to detect suspicious behavior or loitering can encode stereotypes about who counts as threatening and intensify surveillance in already overpoliced neighborhoods (Wahyono et al., 2023). When these outputs are treated as neutral indicators of risk, they can legitimize discriminatory practices and normalize continuous monitoring of communities that social work aims to support.

Yet empirical evaluations also show that human oversight can mitigate these effects. In a mixed-methods study of the Allegheny Family Screening Tool, Cheng and colleagues compared actual worker decisions with those that would have been made if the algorithm alone had determined which referrals to investigate. They found that the algorithm, by itself, would have produced an approximately 18 percentage-point gap in screen time between Black and White children. In contrast, worker decisions using the tool resulted in a gap of about seven percentage points,

because workers frequently overrode high-risk scores for Black families when those scores conflicted with their holistic assessment of the case (Cheng et al., 2022; Stapleton et al., 2022). Similar analyses of Allegheny County's implementation indicate that the introduction of the screening tool, together with continued human discretion, was associated with reductions in Black-White disparities in screening, case openings, and short-term home removals, even though supervisors retained final authority over decisions (Goldhaber-Fiebert & Prince, 2019; Rittenhouse et al., 2023). These cases underscore the importance of human-in-the-loop decision-making rather than fully automated determinations. More broadly, achieving fairness requires systematic auditing, the use of bias detection and mitigation metrics, transparency about model assumptions, and the involvement of affected communities in the design and governance of AI systems (Funda, 2025). Ethical AI in social work must therefore be grounded in participatory processes, culturally responsive data practices, and explicit commitments to equity, ensuring that computational tools support, rather than undermine, social justice aims.

3.3 Epistemological Challenges

3.3.1 Shifts in Knowledge Production and Validation

AI technologies are reshaping how knowledge is generated, interpreted, and validated in social work research (Chen et al., 2024b). Automated literature retrieval, predictive risk scoring, and algorithmic pattern recognition can create the impression that data "speak for themselves," encouraging an overly empiricist orientation in which statistical associations are treated as objective facts. Yet AI outputs inevitably reflect assumptions baked into training data, modeling choices, and system design. Overreliance on algorithmic recommendations risks misinterpretation or misclassification, especially when contextual nuance or experiential knowledge is overlooked (Cross et al., 2024). AI has also altered research workflows across disciplines, expanding analytical capacity while introducing new vulnerabilities related to error propagation, academic misconduct, and uncritical acceptance of machine-generated findings (Chakravorti et al., 2025). These shifts highlight the need for AI-generated insights to be situated within theoretical frameworks, interpreted through professional judgment, and subjected to reflexive critique rather than treated as neutral or authoritative knowledge claims.

3.3.2 The "Black-Box" Problem

The opacity of many AI models raises not only practical concerns but also more profound epistemological questions about the kind of knowledge these systems produce (Greif, 2022). While all statistical models involve abstraction, high-performing machine learning and deep learning architectures are distinct in both scale and

structure. They often contain millions or even billions of parameters, arranged across many layers of nonlinear transformations. As a result, there is no straightforward way to trace how a particular input feature flows through the network or how combinations of features yield a given output (Chang & Chen, 2024; Søgaaard, 2024). This structural complexity makes it difficult for researchers, practitioners, and affected communities to understand why a prediction was made, even when the model's overall accuracy is high.

By contrast, more traditional models such as linear regression, logistic regression, or shallow decision trees provide explicit and relatively stable mappings between inputs and outputs. Coefficients, splits, and rules can be inspected, debated, and related to substantive theory. In complex neural networks, representations are distributed across many units and layers. Individual parameters lack clear meanings, and small changes in inputs can interact in nonintuitive ways (Lange et al., 2022; Xu et al., 2025b). Training procedures are also probabilistic and path dependent, influenced by random initialization, stochastic optimization, and subtle properties of the training data. Two models trained on the same dataset may yield slightly different internal representations and decision boundaries, even if their aggregate performance appears similar (Benzing et al., 2022; Kwok et al., 2025). These features of modern ML and DL make it harder to provide the mechanistic or theory-linked explanations that social work typically expects when justifying interventions or policies.

In social work contexts, the black-box problem is therefore not only about a lack of technical transparency but about who can legitimately interrogate and contest algorithmic reasoning (Busuioc et al., 2022; Hollanek, 2020). When frontline workers are asked to act on predictions from models they cannot explain, they bear responsibility for decisions they cannot articulate to clients, supervisors, or courts. When communities are subject to surveillance or triage by opaque systems, they lack the information needed to challenge potential errors, embedded biases, or harmful feedback loops. Opacity also increases the risk that vulnerabilities in system design, privacy practices, or third-party data flows remain hidden, since even technically sophisticated users may struggle to understand how information is combined and propagated.

To uphold epistemic integrity and institutional accountability, AI used in social research and practice must be paired with forms of transparency meaningful to non-experts. This includes open documentation of training data and model assumptions, explainability tools that reveal how inputs influence outputs at both individual and group levels, and governance arrangements that guarantee a right to explanation and contestation for people whose lives are shaped by algorithmic findings. Without such measures, black-box models threaten to shift social work away from reasoned, accountable judgment toward forms of decision-making that are difficult to justify, scrutinize, or reform.

3.3.3 Reflexivity and Positionality in Human-AI Collaboration

Reflexivity, the critical examination of one's assumptions, standpoint, and interpretive practices, is central to social work scholarship (Rogers & Brown, 2023). AI complicates this process by introducing new forms of cognitive bias and power asymmetry into the production of knowledge. Researchers may inadvertently anchor their interpretations to an initial AI output, overlook omissions, or ascribe unwarranted authority to machine-generated text (Rastogi et al., 2020). These dynamics require heightened reflexive awareness, including careful interrogation of data sources, questioning of model limitations, and triangulation of AI-generated results with independent evidence. Human-AI collaboration in qualitative and conceptual research must treat algorithmic classifications and summaries as provisional, subject to expert review and contextual judgment. Interpretability frameworks underscore that explanations are co-constructed between the system and the user, shaped by the user's positionality, expertise, and expectations (Yang & Ma, 2025). Maintaining epistemic rigor, therefore, requires attention to how human interpretation, institutional context, and structural biases embedded within AI systems jointly shape decisions and knowledge claims.

4 Future Directions

4.1 *Human-AI Collaboration in Research Workflows*

Looking ahead, AI is likely to function as a collaborative partner rather than a replacement for social work researchers. Future research workflows will increasingly rely on systems that augment professional judgment by streamlining data processing, generating preliminary analyses, and assisting with literature synthesis, while leaving interpretation, contextualization, and ethical decision-making to trained professionals. Human review remains essential: examples from risk assessment practice illustrate that skilled practitioners can identify and correct algorithmic biases, significantly reducing disparities in automated recommendations. To support effective collaboration, research environments will need to incorporate AI literacy training, develop interfaces that facilitate the critical examination of model outputs, and design mechanisms to ensure that machine-generated insights remain transparent, auditable, and open to contestation. In this vision of human-AI partnership, the role of the researcher expands from data analyst to orchestrator of hybrid workflows that combine computational efficiency with relational expertise and ethical judgment.

4.2 Epistemological Renewal and Theoretical Integration

AI's ability to detect cross-cutting patterns in large, heterogeneous datasets invites a reassessment of how social problems are conceptualized. Predictive models may reinforce existing theoretical accounts, reveal discrepancies between assumptions and empirical patterns, or challenge dominant constructs altogether. Such insights create opportunities for scholars to interrogate which variables are being optimized, what forms of harm or resilience are foregrounded, and how structural inequalities shape the datasets that models learn from. At the same time, AI-derived findings must not be treated as definitive: cognitive biases such as anchoring can lead researchers to overvalue the first model output they encounter, while the absence of contextual nuance may obscure mechanisms that matter for theory. Epistemological renewal will therefore require integrating AI-generated evidence with critical, interpretive, and values-based traditions in social work. This means treating AI as a tool for provoking theoretical reflection rather than replacing it—one that can help refine constructs, test conceptual boundaries, and illuminate overlooked dimensions of social experience.

4.3 Interdisciplinary Synthesis and Policy Modeling

The future of AI-assisted social work research depends on sustained collaboration across disciplines. Social problems cut across psychological, technological, legal, and policy domains, and effective AI systems must therefore be co-designed by diverse teams that include social scientists, engineers, ethicists, and community stakeholders. Interdisciplinary collaboration can support the development of fairness and transparency standards, improve the cultural responsiveness of datasets, and ensure that AI models align with human rights and social justice principles. These partnerships are particularly valuable in policy modeling, where AI can simulate the consequences of interventions, explore counterfactual scenarios, and integrate evidence from multiple sectors, including health, education, housing, and legal systems. Coordinated efforts among researchers, practitioners, administrators, and affected communities will be essential for building AI systems that not only advance analytical rigor but also reinforce social work's commitments to dignity, inclusion, and equitable outcomes. Such synthesis offers a pathway toward robust, ethically grounded evidence that can inform policy, strengthen service systems, and deepen the field's theoretical foundations.

5 Conclusion

AI has become an increasingly versatile methodological partner in social work research, extending the field's capacity for descriptive, predictive, experimental, and conceptual inquiry. Techniques such as data mining, NLP, machine learning, socially assistive robotics, computer vision, and LLMs enable analysis and synthesis at scales and granularities previously unattainable. Yet the value of these tools is contingent rather than inherent. Their contributions depend on a transparent research design, equitable data governance, and an epistemic posture that treats AI outputs as provisional insights to be interpreted, rather than definitive truths.

Methodologically, AI is most powerful when embedded within rigorous mixed-methods frameworks that ensure clear data provenance, reproducibility, interpretability, and appropriate model validation. Ethically, its use requires robust safeguards to protect privacy, provide informed consent, promote fairness, and ensure human oversight to mitigate potential harms and support accountability, particularly in settings involving vulnerable populations or sensitive service systems. Epistemologically, AI should function as a catalyst for reflection rather than a substitute for human judgment, encouraging researchers to interrogate assumptions, examine sources of bias, and engage critically with how knowledge is produced through human-machine collaboration.

Looking forward, three priorities stand out. First, advancing human-AI collaboration that enhances rather than replaces the intellectual and relational foundations of social inquiry. Second, pursuing epistemological renewal that integrates computational evidence with theoretical traditions and values central to social work. Third, fostering interdisciplinary collaboration to build transparent, responsible, and equity-centered research infrastructures. When guided by methodological rigor, ethical integrity, and the profession's enduring commitments to dignity, justice, and human rights, AI has the potential to help social work ask more profound questions, generate more actionable insights, and design more just and responsive interventions.

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Part V
Ethical Considerations and Future
Directions